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How earthquakes occur

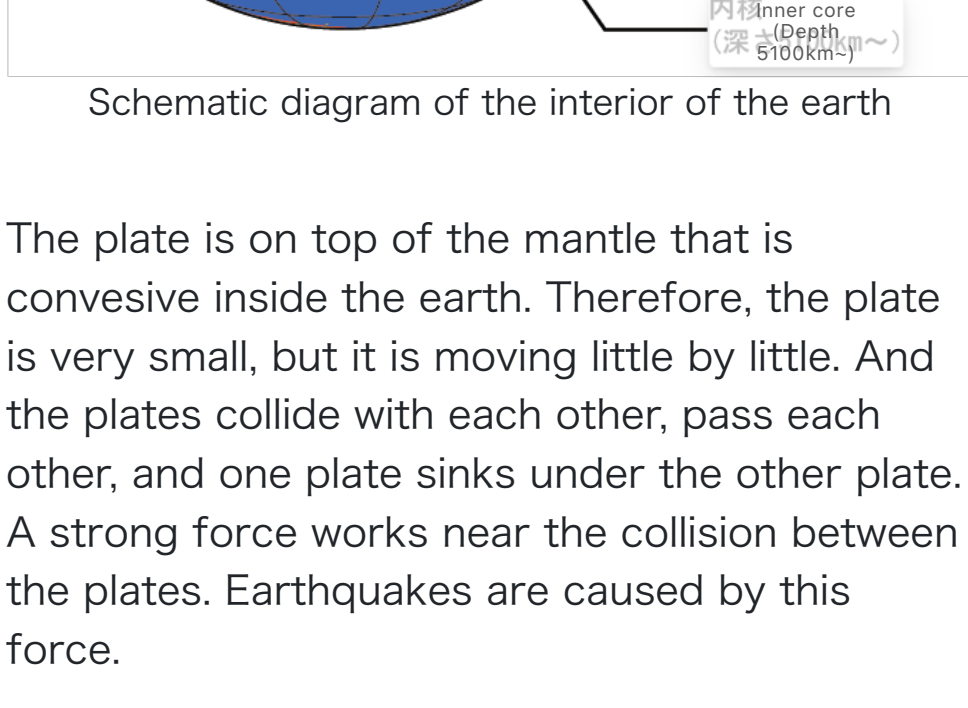
How earthquakes occur

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Why do earthquakes occur? -Plate tectonics-

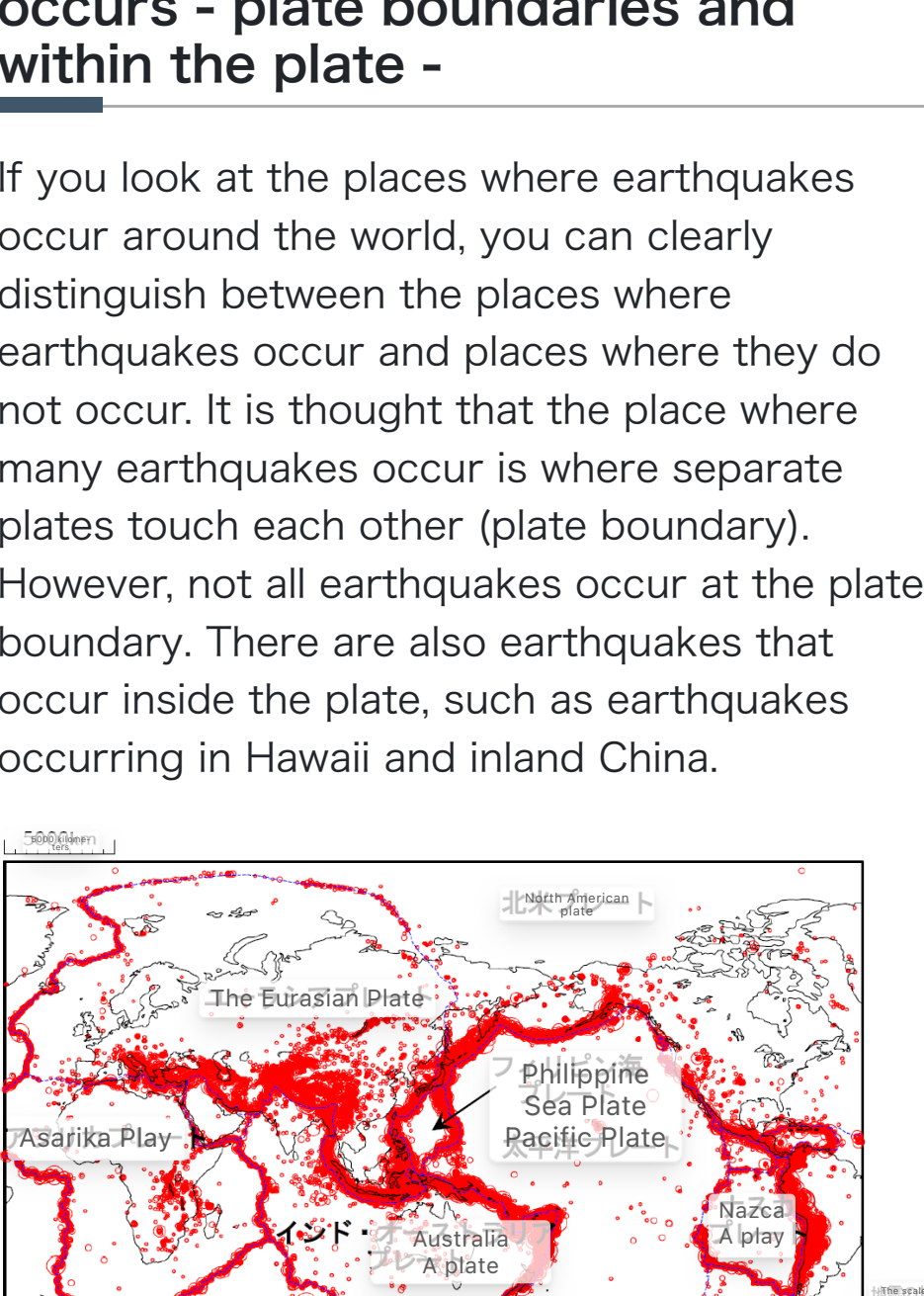
Earthquakes are a phenomenon caused by the "shift" of the bedrock that occurs underground. Then, why does this phenomenon occur? If some kind of force is applied to a hard object and it can't stand it, it will crack. In the same way underground, there is force on the bedrock, and when it can no longer withstand it, an earthquake occurs (the bedrock shifts).

Then, why is there power in the underground? This is explained by the theory of "plate tectonics". It is believed that the earth has a layer structure from the center to the nucleus (inner core, outer core), mantle (lower mantle, upper mantle), and crust. Among them, the "crust" and the part close to the crust of the upper mantle is a hard plate-shaped bedrock, which is called a "plate". The surface of the earth is covered with more than a dozen plates.



Schematic diagram of the interior of the earth

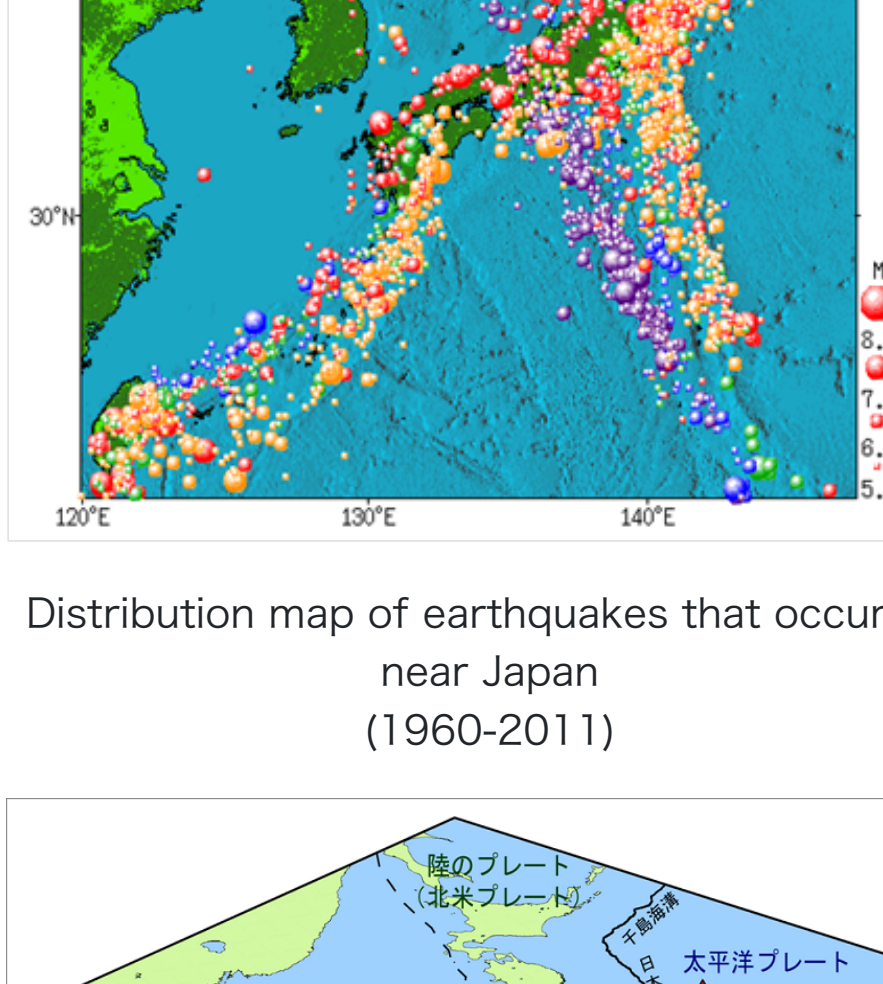
The plate is on top of the mantle that is convective inside the earth. Therefore, the plate is very small, but it is moving little by little. And the plates collide with each other, pass each other, and one plate sinks under the other plate. A strong force works near the collision between the plates. Earthquakes are caused by this force.



Schematic diagram of plate movement

The place where the earthquake occurs - plate boundaries and within the plate -

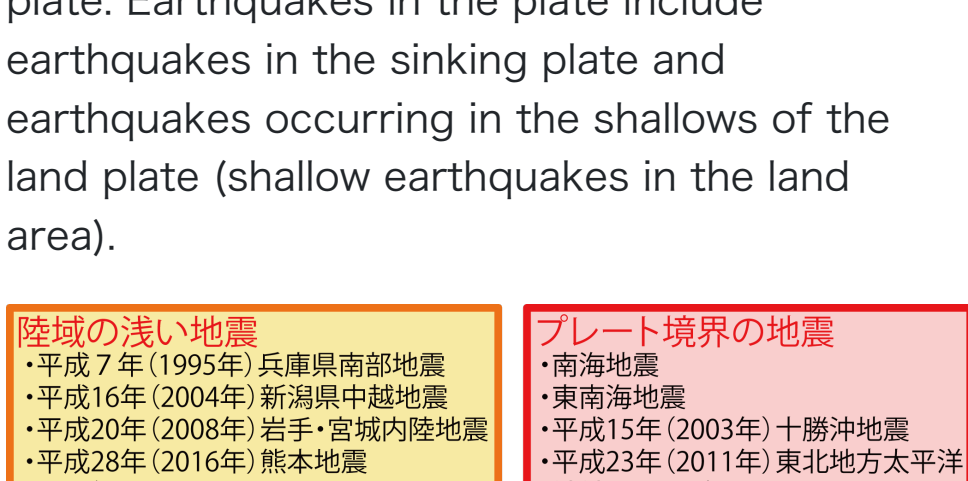
If you look at the places where earthquakes occur around the world, you can clearly distinguish between the places where earthquakes occur and places where they do not occur. It is thought that the place where many earthquakes occur is where separate plates touch each other (plate boundary). However, not all earthquakes occur at the plate boundary. There are also earthquakes that occur inside the plate, such as earthquakes occurring in Hawaii and inland China.



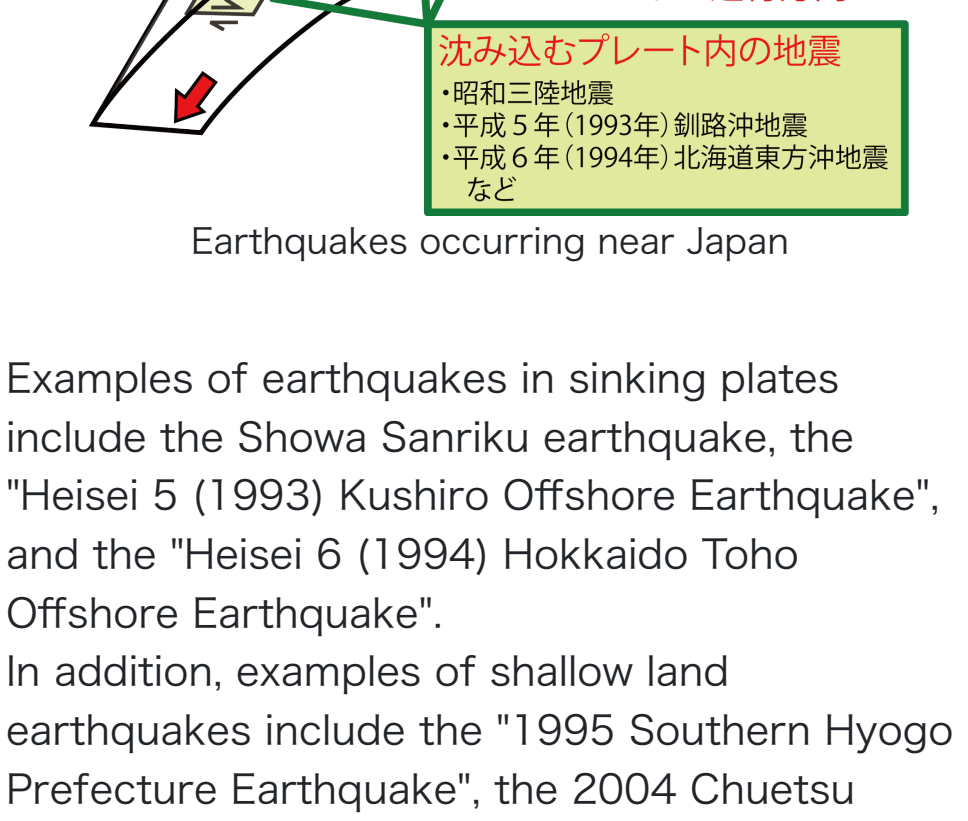
Major plates and earthquake distributions in the world

Places where earthquakes occur around Japan

Around Japan, the Pacific Plate and the Philippine Sea Plate, which are sea plates, are moving towards the land plate (North American Plate and Eurasian Plate) at a rate of several cm per year, and are sinking under the land plate. For this reason, the area around Japan is affected by complex forces due to multiple plates, making it one of the world's most earthquake-prone areas.



Distribution map of earthquakes that occurred near Japan (1960-2011)



Schematic diagram of a plate near Japan

Around Japan, when the sea plate sinks, the land plate is dragged underground. It is the earthquake at the plate boundary that occurs so that the land plate can no longer withstand the drag and bounces up.

Examples of earthquakes at the plate boundary include the Nankai earthquake, the Tonan Sea earthquake, the "2003 Tokachi Oki Earthquake", and the "Heisei 23 (2011) Tohoku Pacific Ocean Earthquake".

An earthquake that occurs when force is applied inside the plate is an earthquake in the plate. Earthquakes in the plate include earthquakes in the sinking plate and earthquakes occurring in the shallows of the land plate (shallow earthquakes in the land area).

陸域の浅い地震 ・平成7年(1995年)兵庫県南部地震 ・平成16年(2004年)新潟県中越地震 ・平成20年(2008年)岩手・宮城内陸地震 ・平成28年(2016年)熊本地震 など	プレート境界の地震 ・南海地震 ・東南海地震 ・平成15年(2003年)十勝沖地震 ・平成23年(2011年)東北地方太平洋沖地震 など
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沈み込むプレート内の地震 ・昭和三陸地震 ・平成5年(1993年)釧路沖地震 ・平成6年(1994年)北海道東方沖地震 など

Earthquakes occurring near Japan

Examples of earthquakes in sinking plates include the Showa Sanriku earthquake, the "Heisei 5 (1993) Kushiro Offshore Earthquake", and the "Heisei 6 (1994) Hokkaido Toho Offshore Earthquake".

In addition, examples of shallow land earthquakes include the "1995 Southern Hyogo Prefecture Earthquake", the 2004 Chuetsu Earthquake in Niigata Prefecture, the 2008 Iwate-Miyagi Inland Earthquake", and the "2016 (2016) Bear There is a "main earthquake".

Shallow land earthquakes are often smaller than earthquakes that occur at plate boundaries, but because they occur close to human habitation areas, they can involve great damage.

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